```
krishnasree08@krishnasree08:~$ sudo apt update
sudo apt install build-essential nano -y
mkdir -p ~/ossp_session7_8
cd ~/ossp_session7_8
[sudo] password for krishnasree08:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
138 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
nano is already the newest version (7.2-2ubuntu0.2).
nano set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 138 not upgraded.
krishnasree08@krishnasree08:~/ossp_session7_8$ gcc -Wall -Wextra waitpid_demo.c -o waitpid_demo
./waitpid_demo
cc1: fatal error: waitpid_demo.c: No such file or directory
compilation terminated.
-bash: ./waitpid_demo: No such file or directory
krishnasree08@krishnasree08:~/ossp_session7_8$ cd ~/ossp_session7_8
pwd
ls -l
/home/krishnasree08/ossp_session7_8
total 0
krishnasree08@krishnasree08:~/ossp_session7_8$ nano waitpid_demo.c
krishnasree08@krishnasree08:~/ossp_session7_8$ gcc -Wall -Wextra waitpid_demo.c -o waitpid_demo
./waitpid_demo
Parent waiting for child process...
Child process is running. PID = 778
Child process completed.
krishnasree08@krishnasree08:~/ossp_session7_8$ cd ~/ossp_session7_8
nano path_resolution.c
krishnasree08@krishnasree08:~/ossp_session7_8$ gcc -Wall -Wextra path_resolution.c -o path_resolution
./path_resolution ls
Executable found: /usr/bin/ls
```

```
krishnasree08@krishnasree08:~/ossp_session7_8$ gcc -Wall -Wextra path_res
./path_resolution ls
Executable found: /usr/bin/ls
krishnasree08@krishnasree08:~/ossp_session7_8$ ./path_resolution abcxyz
Command not found: abcxyz
krishnasree08@krishnasree08:~/ossp_session7_8$
```

Result:

The programs in OSSP Sessions 7 and 8 demonstrate process synchronization using waitpid(), searching for executable files through the PATH environment variable, and expanding environment-variable references. The programs illustrate parent–child process coordination, executable permission checking, environment-variable retrieval, and handling of undefined variables. The first three programs were compiled and executed successfully, subject to confirmation of the actual terminal outputs.